Python Programming – Problem Statements

Course: Introduction to Python
Assignment: Practice Problems
Total Questions: 10

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1. Even or Odd Number

Write a Python program that takes an integer as input from the user and determines whether the number is **even or odd**.

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2. Largest of Three Numbers

Write a Python program that accepts **three numbers from the user** and prints the **largest number** among them.

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3. Factorial of a Number

Write a Python program to **calculate the factorial of a given number using a loop**.

Example:
Factorial of 5 = 5 × 4 × 3 × 2 × 1 = 120

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4. Fibonacci Series

Write a Python program to print the **Fibonacci sequence up to n terms**.

Example:
Input: 6
Output: 0, 1, 1, 2, 3, 5

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5. Prime Number Check

Write a Python program to check whether a given number is a **prime number** or not.

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6. Sum of Digits

Write a Python program to **calculate the sum of digits** of a given number.

Example:
Input: 1234
Output: 10

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7. Reverse a Number

Write a Python program to **reverse the digits of a given number**.

Example:
Input: 1234
Output: 4321

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8. Palindrome Number

Write a Python program to check whether a number is a **palindrome** (a number that reads the same forwards and backwards).

Example:
121 → Palindrome
123 → Not Palindrome

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9. Count Vowels in a String

Write a Python program that takes a **string as input** and counts the number of **vowels (a, e, i, o, u)** present in it.

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10. Simple Calculator

Write a Python program to create a **simple calculator** that performs the following operations based on user choice:

- Addition
- Subtraction
- Multiplication
- Division